ENGS 106 - Coding 3C Write-Up:

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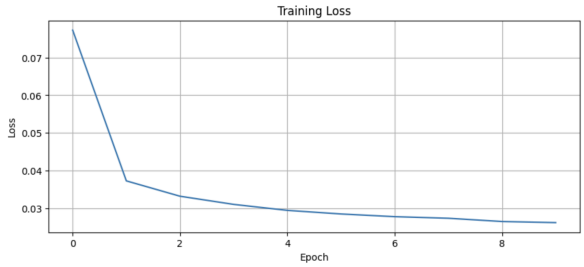
a. Description of the diffusion model and its key componentsI implemented a simple DDPM to generate MNIST digits using the provided skeleton code. It relies on a U-Net backbone. The core pieces are a noise scheduler that handles the 1000-step forward diffusion, sinusoidal embeddings to track timesteps, the U-Net to predict added noise, and a sampling loop that runs reverse DDPM updates to denoise random noise back into digits.

b. Screenshots of generated images and training loss curve



Training Summary: The model trained for 10 epochs (batch size 128, learning rate 2e-4). The total trainable parameters were 1,011,969. Training loss steadily decreased from 0.0773 (Epoch 1) to 0.0262 (Epoch 10), showing that the model successfully learned the denoising objective.

c. Reflection on takeaways This exercise made the forward and reverse diffusion math concrete for me in code. Implementing the sampling process provided a much clearer understanding of how data generation actually happens from iterative denoising.